

The RGCS Crystal Application, Version 3: Anisotropic Propagation, the Node Menu, and the Optical Probe Layer

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Abstract

Version 3 preserves every validated v2 result (geometry, resonance ladder, coupled modes, loading, coherence, uncertainty — all frozen) and adds three tested layers: anisotropic elastic propagation via the Christoffel eigenproblem *RSCS-O.17*, which resolves the v2 scalar wave-speed hypothesis into a measured-orientation model; an explicit eight-entry *node menu* that treats “the eye” as a modeling choice to be measured, not a fact; and an optical probe layer (quartz indices, photoelastic response, ray/path model) with pre-registered claims whose directional entries are expected *nulls*. Experimental status is stated plainly: the physical hypotheses of the programme remain unconfirmed; what this application provides is a reproducible model stack and a falsification plan.

1 Forward hierarchy

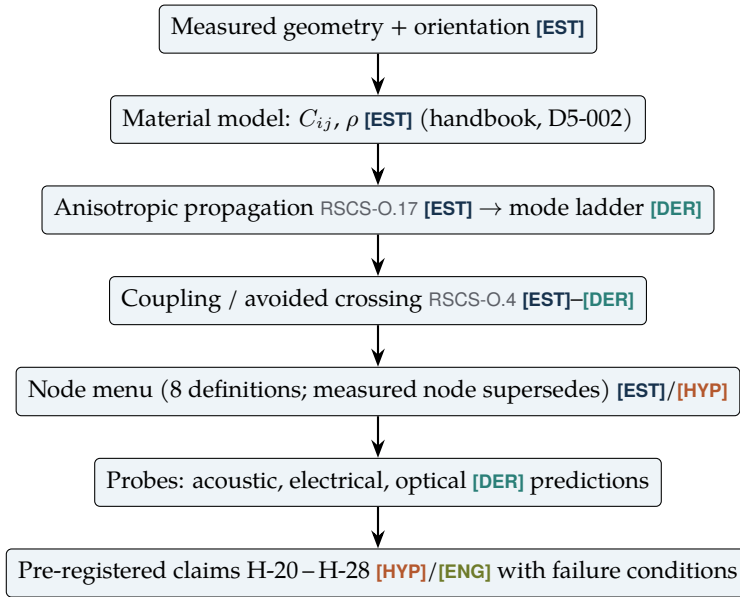


Figure 1: Forward hierarchy: measurement feeds models; models feed predictions; predictions feed pre-registered falsification rows. Nothing flows backward from source lore into the model without a classification boundary.

2 Anisotropic propagation

v2 used one scalar effective speed $v_L = 6310 \pm 5\%$ m/s *RGCS-M.10*, explicitly a *[HYP]* that a single scalar suffices. v3 resolves it: the Christoffel eigenproblem

$$\Gamma_{ik} = c_{ijkl} n_j n_l, \quad \Gamma \mathbf{u} = \rho v^2 \mathbf{u}, \quad \text{RSCS-O.17 [EST]} \quad (1)$$

with the handbook α -quartz constants [1, 2] ($\rho = 2648 \text{ kg/m}^3$, D5-002) yields the quasi-longitudinal and quasi-shear speeds along any direction:

Table 1: α -quartz bulk wave speeds along the crystal axes (m/s), Christoffel eigenproblem RSCS-O.17 with the handbook constants (D5-002).

Axis	v_{qL}	v_{qS1}	v_{qS2}
X	5719	5097	3307
Y	5993	4315	3884
Z	6330	4672	4672

Along Z, $v_{qL} = 6330 \text{ m/s}$ — 0.3% from the v2 default, inside its band [5994, 6626] m/s — and along X, $v_{qL} = 5719 \text{ m/s}$. The v2 $\pm 5\%$ band is thereby *explained* as the physical X–Z anisotropy spread rather than an undeclared systematic (Figure 2). Conservative extension is machine-tested.

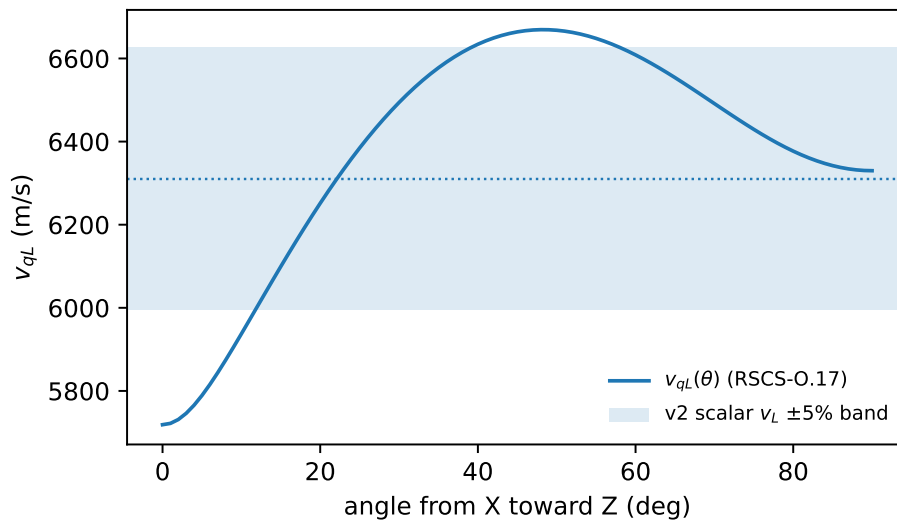


Figure 2: v_{qL} versus propagation direction from X toward Z RSCS-O.17. The shaded region is the frozen v2 scalar band; the anisotropic curve stays within it, recovering the scalar model as the orientation-unknown special case (model-selection rule Section 5).

3 The node menu

“The eye” of the historical source material is treated as a *menu of eight competing definitions*, each classified, none presumed: (1) metric centre [EST], (2) prismatic-shaft prior [DER], (3) measured vibration node [EST] (measured; supersedes all others, v2 H-07), (4) electrical node [HYP] (H-24), (5) optical path/phase feature [HYP] (H-25), (6) maximal multimode overlap [HYP] (H-26), (7) coupling-integral maximum [HYP] (H-27), (8) phase singularity/saddle [HYP] (H-28). Rows H-24–H-28 each carry an observable, controls, and a pre-registered failure condition; a definition that fails is retired, not reinterpreted.

4 Optical probe layer

Quartz is uniaxial positive: $n_o = 1.5443$, $n_e = 1.5534$ ($\Delta n = 0.0091$ at 589.3 nm) [3]. Three conventional mechanisms give [DER] predictions:

1. **Photoelastic phase modulation:** $\Delta n = -\frac{1}{2}n^3 p S$ [4]; with p_{11} and a typical acoustic strain $S = 10^{-7}$, $\Delta n = -2.95e-08$ — small but standard interferometric territory.

2. **Birefringence modulation** between crossed polarizers.
3. **Photothermal absorption** — an artifact channel, bounded by a matched-power off-node control (H-22 [\[ENGI\]](#)). The acousto-optic figure of merit $M_2 = n^6 p^2 / (\rho v^3) = 7.01e - 16 \text{ s}^3/\text{kg}$ (computed with the anisotropic X-axis speed) is 10^2 – $10^3\times$ below dedicated acousto-optic materials: an honest sensitivity bound. A straight-ray entry-facet model answers the geometric question — an optical path *can* address any interior target, with the geometric centre, predicted node, and measured node kept as three distinct locations. Directionality: an unbiased passive quartz path is reciprocal; H-21 and H-23 are pre-registered *null* expectations (D6-003), and any surviving asymmetry would first face a conventional-mechanism audit [5, 6].

5 Model selection: earning complexity

Each layer must beat a simpler baseline before it is used: the anisotropic model applies only when orientation is *measured* (else the scalar band); the compact-coordinate spectrum only if it beats a plain rod on held-out modes (v2 H-01); the coupled-mode model only in the strong-coupling regime; the measured node supersedes every prior. Complexity that does not change a prediction is refused.

6 Environmental and fixture factors

Surface finish, inclusions, electrodes, fixtures, hand loading (v2, frozen), thermal drift of the elastic constants ($\sim 10^{-4}/\text{K}$ [\[DERI\]](#)) and humidity/adsorbate mass loading are declared inputs of the crystal database, each with an uncertainty; the experiment programme’s thermal control branch bounds them.

7 Status

Implemented and machine-tested: anisotropic propagation, quasi-mode taxonomy, optical constants and estimates, ray/path model, crystal database with schema migration, FEA material-card export. Pre-registered and awaiting bench work: H-20–H-28. No physical hypothesis of the programme is confirmed; this application ships the models, the uncertainty discipline, and the falsification plan.

References

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